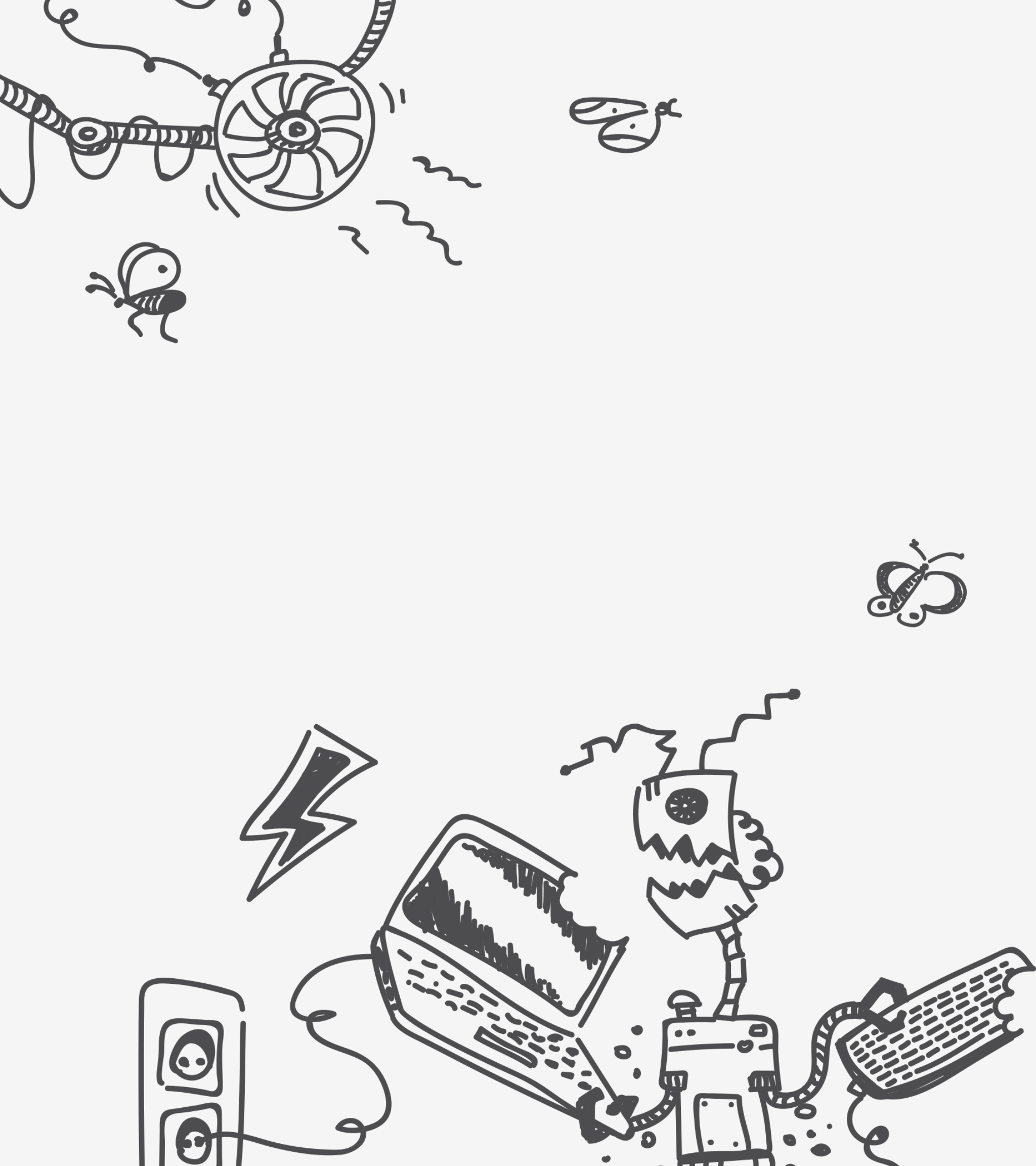
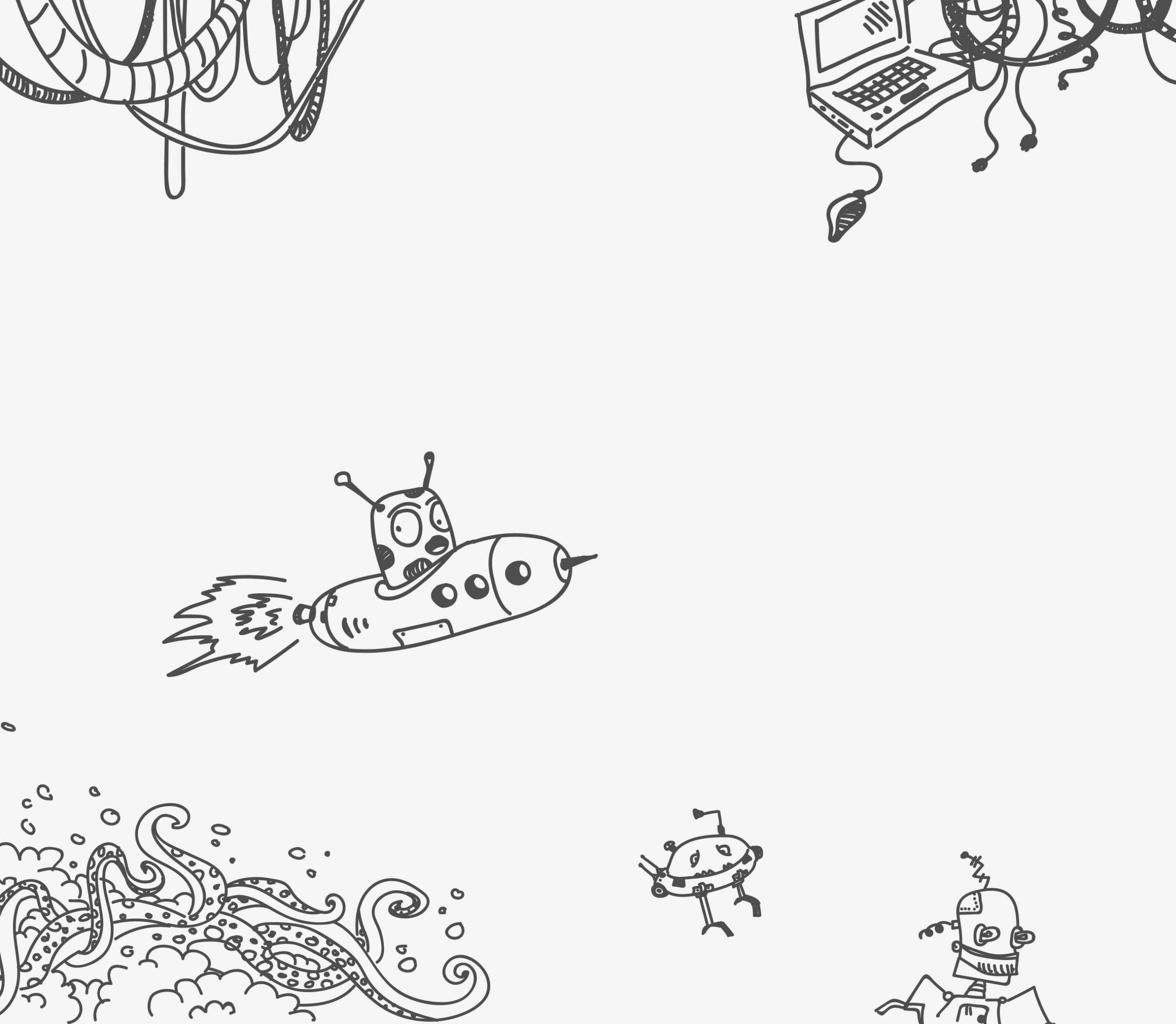




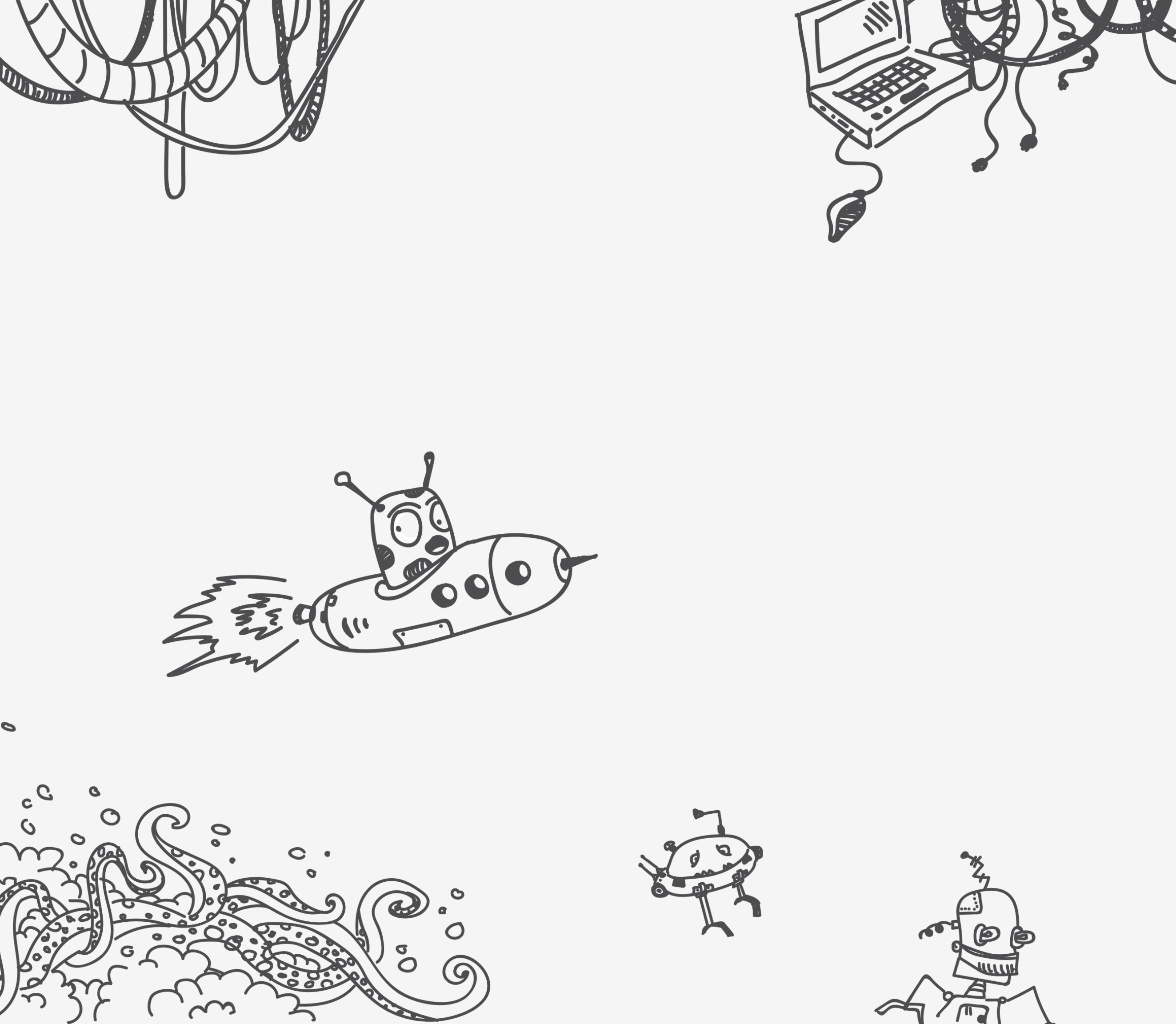
Day 1 summary



Introduction

Introduction

We have reached the end of the first day of learning together on the **Machine Learning** course. We hope it was an effective day, during which you were able to get acquainted with many interesting machine learning concepts. During the course, we touched on regression - that is, a method of modeling relationships between variables with the goal of predicting continuous numerical values (e.g., sales value, product price, credit score, wages, temperature forecast). We can use a number of algorithms to solve a regression problem.

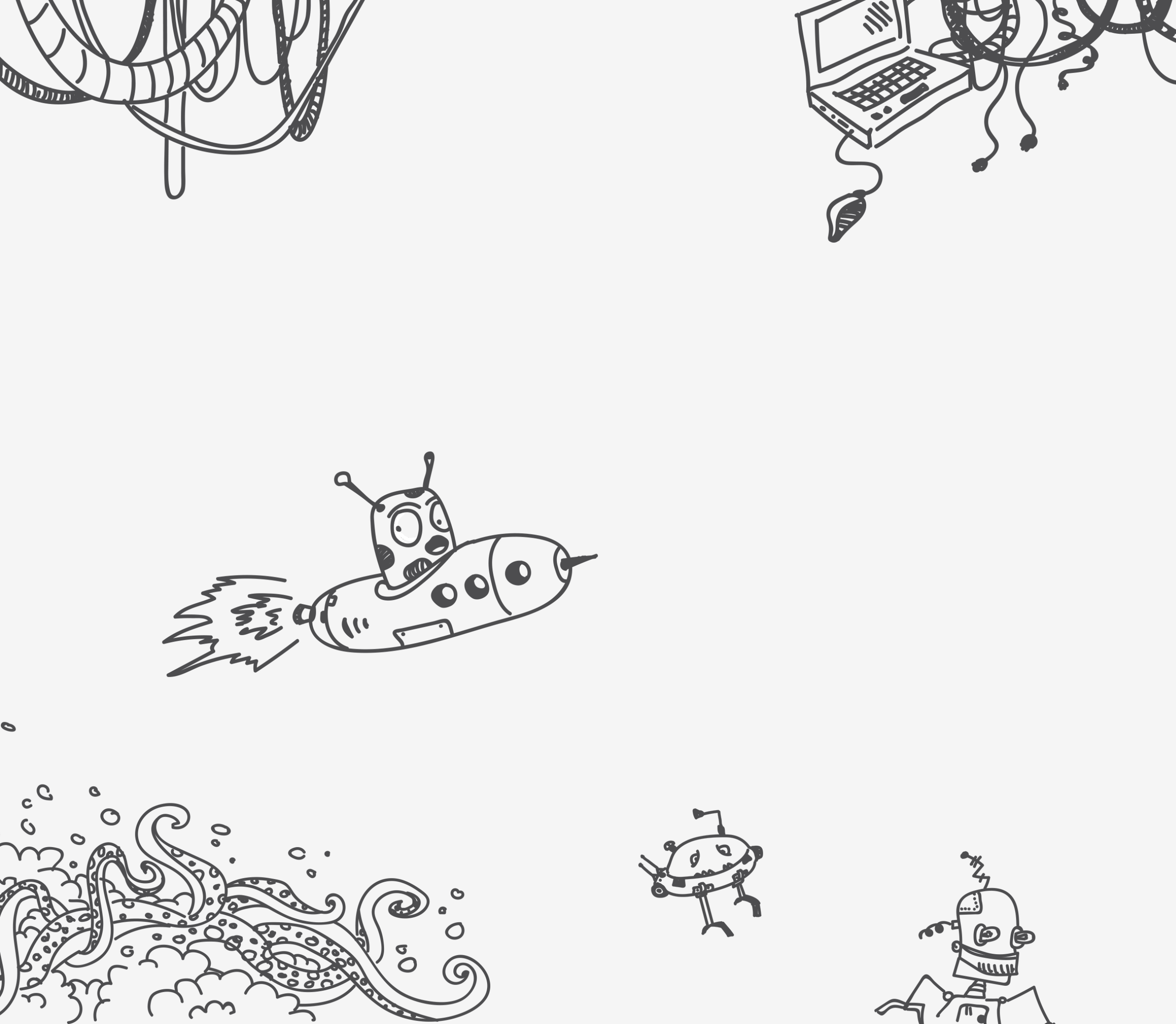


Linear regression

Linear regression

Linear regression is the simplest model that examines the relationship between the independent variables and the dependent variable to predict a continuous value. It assumes that there is a linear relationship between the dependent variable and the other variables. The primary task of the model is to minimize the sum of squares of the errors, which is called the *least squares method*.

Linear regression estimates the values of the regression coefficients (**a** and **b**), which determine the weight that each independent variable has on the value of the dependent variable. Linear regression is relatively easy to interpret - for example, a positive directional coefficient of a straight line suggests that an increase in the independent variable (e.g., earnings) affects an increase in the dependent variable (e.g., credit score). In addition, the linear regression model has a number of tools to evaluate it - *residuals* analysis, multicollinearity analysis, evaluation of the distribution of residuals (their dispersion).

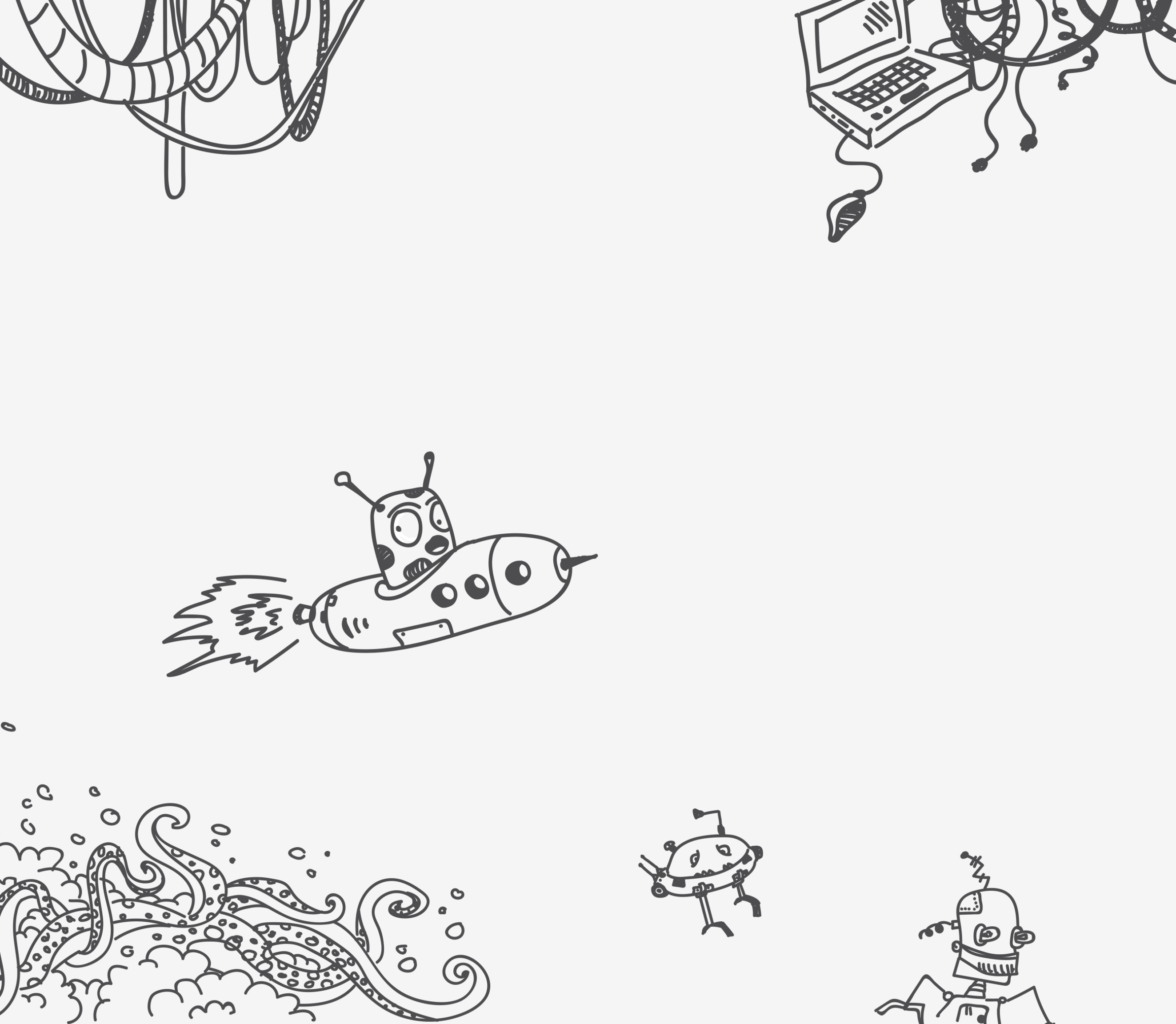


Evaluating the quality of the model

Evaluating the quality of the model

Assuming that we already have a model, we would like to know what level of accuracy it presents. Therefore, we learned about metrics useful in assessing the quality of any regression model.

- The coefficient of determination (**R^2**) is a measure that tells us how well a model fits the data. It takes values from 0 to 1. It measures the percentage of variance in the dependent variable, so it explains how well our problem is solved by the model. There is also an adjusted coefficient of determination, which takes into account the number of independent variables and the number of observations. The adjusted R^2 penalizes the addition of independent variables that have no real effect on improving the quality of the model.
- Mean absolute error (**MAE**) is the average of the absolute differences between the predicted and actual values of the dependent variable. MAE is a metric that is easy to interpret and less sensitive to outliers than MSE.
- Mean squared error (**MSE**) is a popular metric for regression models. It calculates the mean value of the squares of the differences between the predicted values of the dependent variables and the actual values. The smaller the value of this metric, the better the model we have. As the error difference is squared, the MSE is more sensitive to the presence of outliers. MSE tends to take large values (due to being squared), so the more easily interpretable metric is **RMSE**, the root of the mean squared error.



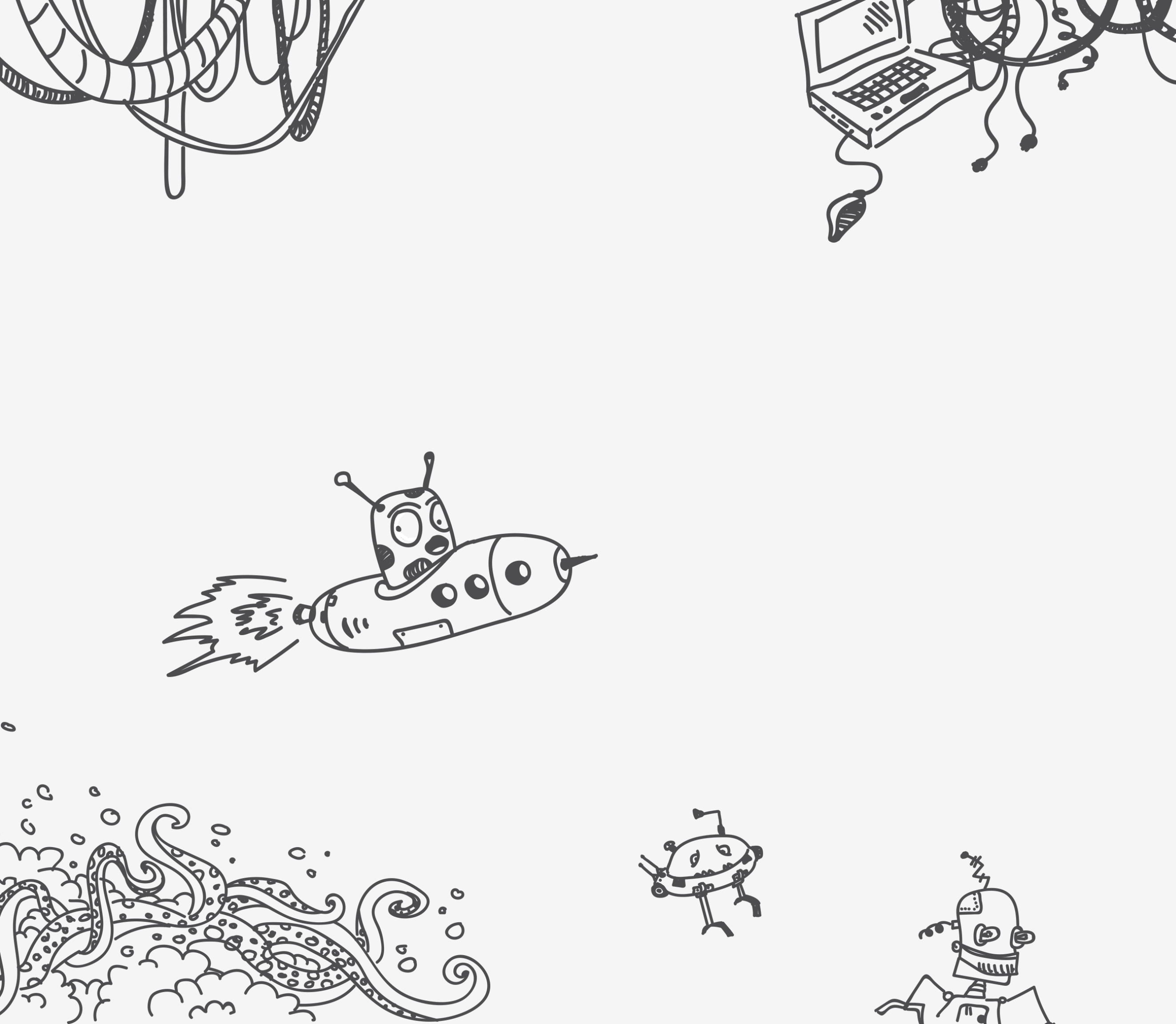
Regularization in linear regression

Regularization in linear regression

Regularization is a technique used in machine learning that aims to control model complexity by introducing additional penalties in the model's cost function. Its goal is to reduce overfitting and improve the model's overall ability to generalize new data. The main idea of regularization is to add an additional component to the cost function that penalizes or limits the values of the model parameters.

L1 regularization (*Lasso*) involves adding a component equal to the sum of the absolute values of the parameter weights multiplied by the regularization strength factor. Lasso regularization helps to select variables that are relevant to the model, zeroing out the coefficient values at irrelevant variables.

L2 regularization (*Ridge*) involves adding a component equal to the sum of the squares of the parameter weights multiplied by a coefficient that determines the strength of the regularization. This type of regularization seeks to reduce the influence of less significant variables (but not exclude them) and reduce the effect of multicollinearity.



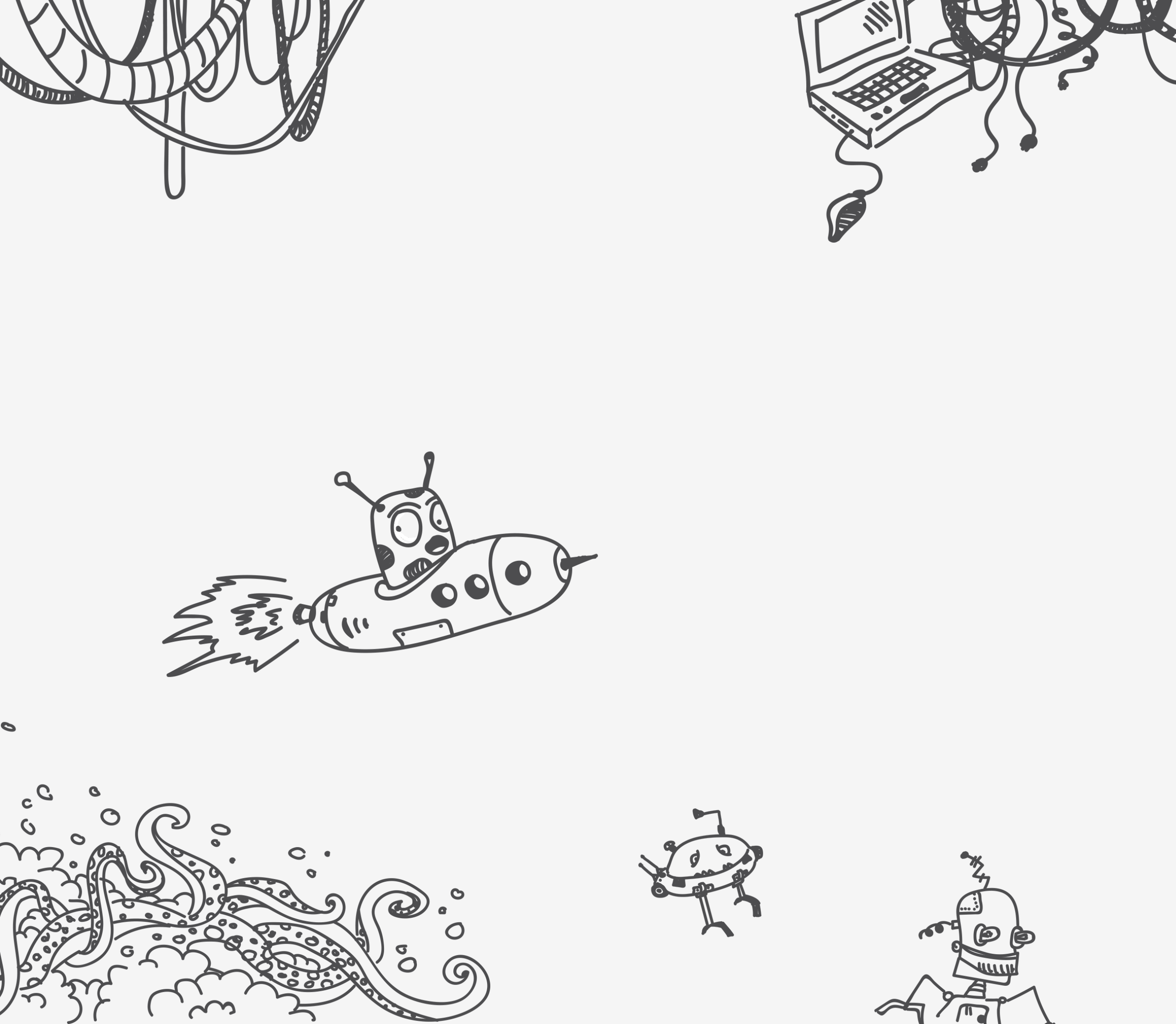
Polynomial regression

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Polynomial regression

Polynomial regression is an extension of linear regression - it allows to model more complex relationships between data. In polynomial regression, polynomials of individual independent variables are used, which makes it possible to effectively model non-linear relationships. Polynomial regression allows modeling curved relationships between variables.

Polynomials can have different degrees of complexity, which is influenced by the degree of the polynomial - a polynomial of 2nd degree is a parabolic curve, a polynomial of 3rd degree is a cubic curve, and a higher value of the degree of the polynomial results in an increasingly complex curve. The greater the degree of the polynomial, the greater the risk of overfitting the model, so you need to pay attention to this when training and testing polynomial regression models.

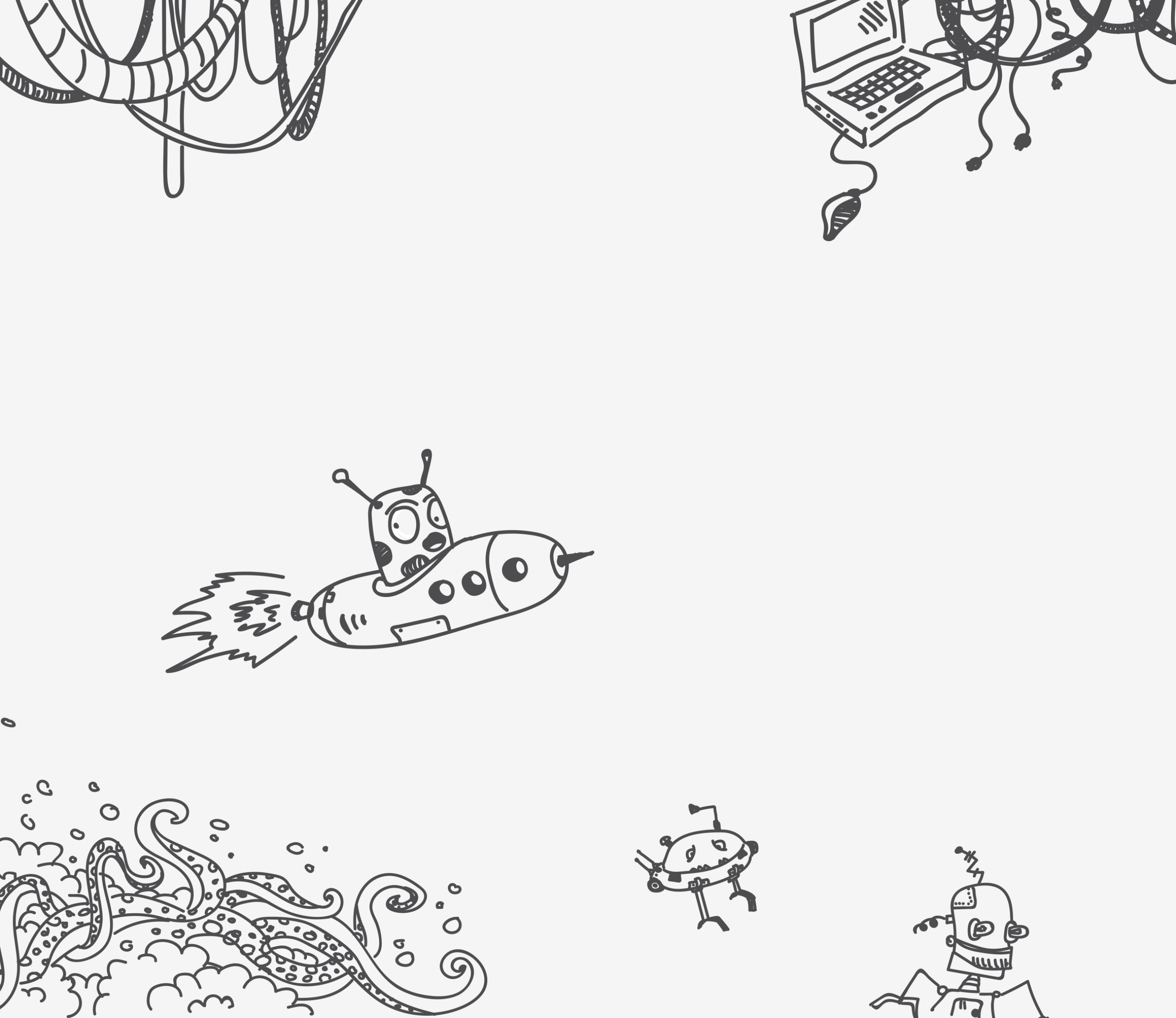


Decision tree in regression

Decision tree in regression

The decision tree is the last algorithm we discussed on the first day of the course. A decision tree consists of nodes where either a division of the data into successive subsets is made or the final decision of the model is made (the final node). The splits performed at the nodes are done in such a way as to maximize the homogeneity of the data at the nodes.

Decision trees have many advantages: they are easily interpreted, can cope with non-linear data, and are not susceptible to different scales in independent variables. In addition, they can cope with data containing outliers. A potential problem can be the possibility of overfitting as the complexity of the tree structure increases. The solution to this is to use ensemble methods such as *bagging* or *boosting*, which we will talk about during the following days of the course.

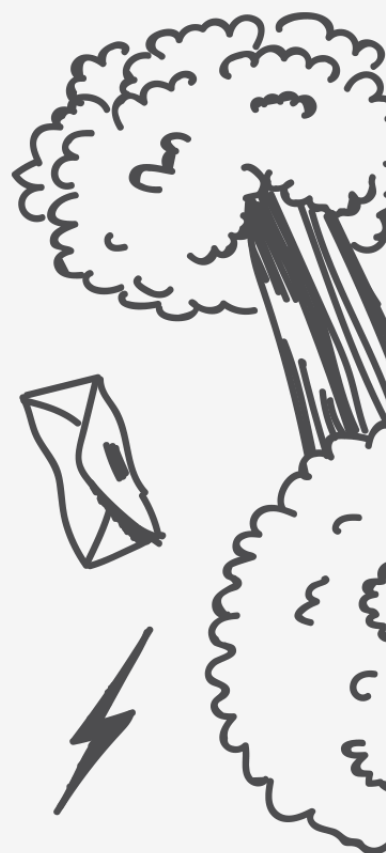
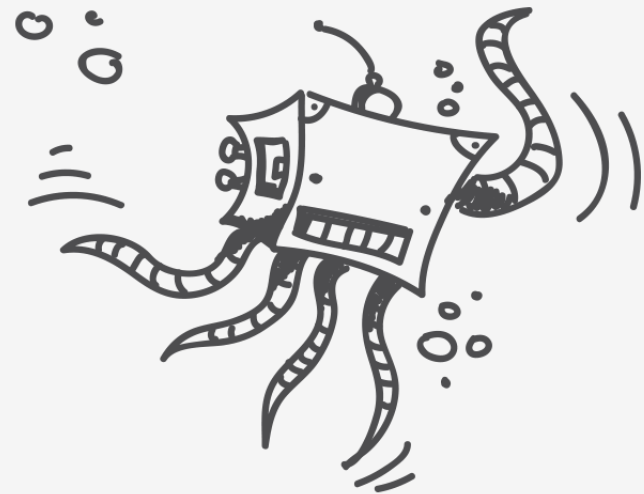
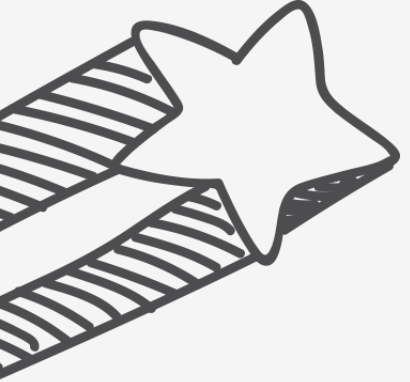


Summary

Summary

As a data scientist, you will often face the problem of regression. Thanks to the knowledge gained during the first day of the course, you get an idea of what regression is all about. You know and understand the most popular algorithms to solve these types of problems. In our materials we have covered the most relevant issues related to each topic. To consolidate the knowledge you have gained, we have prepared a quiz on regression models.

Good luck!



Time
for a quiz